

# W019 Strain Build List

Michael Volk

August 24, 2026

Forty-five strains to construct: 25 double knockouts and 20 triple knockouts. Every one of them is made from a strain that already exists, so no new single knockout is needed and no cross waits on another. Tables 3 and 4 are the build list. Tables 1 and 2 are the strains in hand, and Table 5 is the plate they are all measured on.

Tables generated by `experiments/W019-echo-crispr-array/scripts/build_list_tables.py`

Working notes: `notes/experiments.W019-echo-crispr-array/*.md`

*Section status:* ✓ ready   ■ author-reviewed, keep stable   ✗ not done, agents may edit freely

*Provenance:* ▲ not in the library mirror; verify at the linked source   ● from a review's summary table, not the primary paper

## Contents

|   |                               |   |
|---|-------------------------------|---|
| 1 | Terms ✗                       | 1 |
| 2 | Strains that already exist ✗  | 1 |
| 3 | The 25 doubles to construct ✗ | 2 |
| 4 | The 20 triples to construct ✗ | 3 |
| 5 | The measurement plate ✗       | 4 |

## 1 Terms ✗

**boot SE** The standard error of a strain's mean fitness, from resampling the replicate plates that strain was measured on. It is smaller than *plate SD* because it is the error on a mean rather than the spread of the values that mean was taken over.

**fitness** Colony growth relative to the wild type on the same plate, so 1 is wild-type growth.

**plate SD** The standard deviation of a strain's fitness across the replicate plates it was measured on.

**routes** How many different ways the bench can make a triple from strains that already exist. Every triple is built by crossing one existing double with the remaining single. *routes* 1 means only one of the triple's three pairs is in the freezer, so there is a single way in, and a failed cross blocks that triple until the matching new double from Table 3 is done. *routes* 2 means two of its pairs exist, so a failed cross has an immediate second path.

**tier** How a double was classed when run 4 was designed. **validation** reproduces a measurement someone has already published, **coverage** was added so that later triples would have a parent to be built from, and **novel** had not been measured by anyone.

## 2 Strains that already exist ✗

Twelve singles and thirteen doubles were built and measured in run 4. No triple has been constructed in any round.

**Table 1:** Single knockouts that already exist, all twelve measured in run 4. The three columns under *Ours, run 4* are this lab’s measurement; the two under *Costanzo 2016* are the published single-mutant fitness and its standard error.

| id  | ORF       | common | Ours, run 4 |         |          | Costanzo 2016 |        |
|-----|-----------|--------|-------------|---------|----------|---------------|--------|
|     |           |        | fitness     | boot SE | plate SD | SMF           | SE     |
| s1  | YJR060W   | CBF1   | 0.5348      | 0.0506  | 0.1076   | 0.5900        | 0.1138 |
| s2  | YPL081W   | RPS9A  | 0.9007      | 0.0515  | 0.1105   | 0.9550        | 0.1553 |
| s3  | YDR057W   | YOS9   | 1.2430      | 0.1282  | 0.2690   | 1.0435        | 0.0535 |
| s4  | YPL046C   | ELC1   | 1.2066      | 0.0495  | 0.1052   | 1.0433        | 0.0643 |
| s5  | YBR203W   | COS111 | 1.1292      | 0.0679  | 0.1426   | 1.0370        | 0.0458 |
| s6  | YGL087C   | MMS2   | 1.1628      | 0.0329  | 0.0706   | 0.9960        | 0.0940 |
| s7  | YLR104W   | LCL2   | 1.1748      | 0.0564  | 0.1187   | 1.0322        | 0.0453 |
| s8  | YLL012W   | YEH1   | 1.2604      | 0.0659  | 0.1402   | 0.9925        | 0.0804 |
| s9  | YER079W   | –      | 1.3333      | 0.1504  | 0.3162   | 1.0387        | 0.1173 |
| s10 | YKL033W-A | –      | 0.9251      | 0.0445  | 0.0954   | 1.0327        | 0.1008 |
| s11 | YLR313C   | SPH1   | 1.0882      | 0.1471  | 0.3083   | 0.9843        | 0.0271 |
| s12 | YLR312C-B | –      | 0.8390      | 0.0661  | 0.1397   | 1.0845        | 0.0441 |

**SPH1 is an open question.** s11, SPH1 (*YLR313C*), is a built single that is part of no triple, at a measured fitness of 1.0882 (Table 1). It can ride along on the measurement plate as a near-wild-type control, or it can be dropped. If it rides along, the plate carries 71 strains rather than 70, at the same five wells each.

**Table 2:** Double knockouts that already exist, all thirteen measured in run 4. Column attribution is as in Table 1, with the published double-mutant fitness and its standard deviation under *Costanzo 2016*. *parent* says whether the pair is the starting strain for one of the triples in Table 4. A dash in the Costanzo columns means no one has published the pair.

| id  | pair                | Ours, run 4 |         |          | Costanzo 2016 |        |            |        |
|-----|---------------------|-------------|---------|----------|---------------|--------|------------|--------|
|     |                     | fitness     | boot SE | plate SD | DMF           | SD     | tier       | parent |
| d1  | YDR057W + YPL081W   | 0.6071      | 0.0479  | 0.1029   | 1.0351        | 0.0447 | coverage   | yes    |
| d2  | YPL046C + YPL081W   | 0.9335      | 0.0040  | 0.0084   | –             | –      | novel      | no     |
| d3  | YER079W + YPL081W   | 0.9496      | 0.0556  | 0.1195   | 0.8625        | 0.0975 | validation | no     |
| d4  | YLR312C-B + YPL081W | 0.6480      | 0.0454  | 0.0968   | 1.0165        | 0.0418 | coverage   | yes    |
| d5  | YDR057W + YGL087C   | 0.7884      | 0.0007  | 0.0015   | 1.1372        | 0.0238 | validation | yes    |
| d6  | YDR057W + YER079W   | 0.9403      | 0.0472  | 0.0991   | 1.0963        | 0.0541 | coverage   | no     |
| d7  | YDR057W + YLR312C-B | 0.8895      | 0.0543  | 0.1150   | 1.1783        | 0.0257 | validation | no     |
| d8  | YJR060W + YPL046C   | 0.5649      | 0.0435  | 0.0930   | 0.9663        | 0.0178 | coverage   | yes    |
| d9  | YBR203W + YPL046C   | 0.8882      | 0.0232  | 0.0501   | 1.1179        | 0.0171 | coverage   | yes    |
| d10 | YDR057W + YLL012W   | 0.9315      | 0.0269  | 0.0572   | 0.9995        | 0.0690 | coverage   | yes    |
| d11 | YLL012W + YPL046C   | 0.9087      | 0.0348  | 0.0744   | 1.0251        | 0.0534 | coverage   | yes    |
| d12 | YER079W + YJR060W   | 0.6886      | 0.0650  | 0.1365   | 0.6100        | 0.0183 | validation | no     |
| d13 | YER079W + YLR312C-B | 0.7728      | 0.0514  | 0.1084   | 1.0994        | 0.0393 | coverage   | yes    |

All thirteen doubles go on the measurement plate. The pairs marked **yes** under *parent* are the starting strains for Table 4; the rest are re-measured alongside them, at no cost in construction work and for a second reading on each of those pairs.

**One designed double was never built.** *YJR060W*  $\times$  *YKL033W-A* does not exist: the transformation produced no colonies. Both parent singles exist and were measured. The pair is not part of the build list, and nothing in Tables 3 or 4 depends on it.

### 3 The 25 doubles to construct $\times$

Cross the two singles named in the row. Both parents are already in the freezer (Table 1) and no row depends on another, so all 25 can be started at once. Each row gives the ORF and, where the gene has one, the common name in parentheses.

**Table 3:** The 25 double knockouts to construct, one per row. Each row names the two singles to cross, both of them in Table 1.

| ID  | gene 1           | gene 2          |
|-----|------------------|-----------------|
| D01 | YBR203W (COS111) | YDR057W (YOS9)  |
| D02 | YBR203W (COS111) | YGL087C (MMS2)  |
| D03 | YBR203W (COS111) | YJR060W (CBF1)  |
| D04 | YBR203W (COS111) | YKL033W-A       |
| D05 | YBR203W (COS111) | YLL012W (YEH1)  |
| D06 | YBR203W (COS111) | YLR104W (LCL2)  |
| D07 | YBR203W (COS111) | YLR312C-B       |
| D08 | YBR203W (COS111) | YPL081W (RPS9A) |
| D09 | YDR057W (YOS9)   | YKL033W-A       |
| D10 | YER079W          | YLL012W (YEH1)  |
| D11 | YER079W          | YLR104W (LCL2)  |
| D12 | YGL087C (MMS2)   | YLL012W (YEH1)  |
| D13 | YGL087C (MMS2)   | YLR312C-B       |
| D14 | YGL087C (MMS2)   | YPL046C (ELC1)  |
| D15 | YGL087C (MMS2)   | YPL081W (RPS9A) |
| D16 | YJR060W (CBF1)   | YLL012W (YEH1)  |
| D17 | YJR060W (CBF1)   | YLR104W (LCL2)  |
| D18 | YKL033W-A        | YLL012W (YEH1)  |
| D19 | YKL033W-A        | YLR312C-B       |
| D20 | YKL033W-A        | YPL046C (ELC1)  |
| D21 | YKL033W-A        | YPL081W (RPS9A) |
| D22 | YLL012W (YEH1)   | YLR312C-B       |
| D23 | YLR104W (LCL2)   | YLR312C-B       |
| D24 | YLR104W (LCL2)   | YPL046C (ELC1)  |
| D25 | YLR104W (LCL2)   | YPL081W (RPS9A) |

## 4 The 20 triples to construct $\times$

Cross the existing double in *build from* with the single in *third single*. Start with T01 through T15, which have one route each: a failed cross there blocks the triple until the matching double from Table 3 is finished, so those failures are worth surfacing early. T16 through T20 each have a second route already in hand and can absorb a failed cross without waiting on anything.

**Table 4:** The 20 triple knockouts to construct, in build order. *build from* names an existing double by ORF and *third single* the remaining gene; crossing the two gives the triple. *routes* counts how many different ways the bench can make the triple from strains that already exist, and is defined in Sec. 1.

| ID  | gene 1           | gene 2         | gene 3          | build from          | $\times$ third single | routes |
|-----|------------------|----------------|-----------------|---------------------|-----------------------|--------|
| T01 | YKL033W-A        | YLR312C-B      | YPL081W (RPS9A) | YLR312C-B + YPL081W | YKL033W-A             | 1      |
| T02 | YER079W          | YLL012W (YEH1) | YLR312C-B       | YER079W + YLR312C-B | YLL012W (YEH1)        | 1      |
| T03 | YLR104W (LCL2)   | YLR312C-B      | YPL081W (RPS9A) | YLR312C-B + YPL081W | YLR104W (LCL2)        | 1      |
| T04 | YBR203W (COS111) | YLR312C-B      | YPL081W (RPS9A) | YLR312C-B + YPL081W | YBR203W (COS111)      | 1      |
| T05 | YER079W          | YLR104W (LCL2) | YLR312C-B       | YER079W + YLR312C-B | YLR104W (LCL2)        | 1      |
| T06 | YGL087C (MMS2)   | YLR312C-B      | YPL081W (RPS9A) | YLR312C-B + YPL081W | YGL087C (MMS2)        | 1      |
| T07 | YBR203W (COS111) | YDR057W (YOS9) | YPL081W (RPS9A) | YDR057W + YPL081W   | YBR203W (COS111)      | 1      |
| T08 | YDR057W (YOS9)   | YKL033W-A      | YPL081W (RPS9A) | YDR057W + YPL081W   | YKL033W-A             | 1      |
| T09 | YDR057W (YOS9)   | YKL033W-A      | YLL012W (YEH1)  | YDR057W + YLL012W   | YKL033W-A             | 1      |
| T10 | YGL087C (MMS2)   | YLL012W (YEH1) | YPL046C (ELC1)  | YLL012W + YPL046C   | YGL087C (MMS2)        | 1      |
| T11 | YBR203W (COS111) | YLR104W (LCL2) | YPL046C (ELC1)  | YBR203W + YPL046C   | YLR104W (LCL2)        | 1      |
| T12 | YBR203W (COS111) | YKL033W-A      | YPL046C (ELC1)  | YBR203W + YPL046C   | YKL033W-A             | 1      |
| T13 | YJR060W (CBF1)   | YLR104W (LCL2) | YPL046C (ELC1)  | YJR060W + YPL046C   | YLR104W (LCL2)        | 1      |
| T14 | YBR203W (COS111) | YGL087C (MMS2) | YPL046C (ELC1)  | YBR203W + YPL046C   | YGL087C (MMS2)        | 1      |
| T15 | YBR203W (COS111) | YDR057W (YOS9) | YLL012W (YEH1)  | YDR057W + YLL012W   | YBR203W (COS111)      | 1      |
| T16 | YDR057W (YOS9)   | YGL087C (MMS2) | YPL081W (RPS9A) | YDR057W + YGL087C   | YPL081W (RPS9A)       | 2      |
| T17 | YJR060W (CBF1)   | YLL012W (YEH1) | YPL046C (ELC1)  | YJR060W + YPL046C   | YLL012W (YEH1)        | 2      |
| T18 | YBR203W (COS111) | YLL012W (YEH1) | YPL046C (ELC1)  | YBR203W + YPL046C   | YLL012W (YEH1)        | 2      |
| T19 | YBR203W (COS111) | YJR060W (CBF1) | YPL046C (ELC1)  | YBR203W + YPL046C   | YJR060W (CBF1)        | 2      |
| T20 | YDR057W (YOS9)   | YGL087C (MMS2) | YLL012W (YEH1)  | YDR057W + YGL087C   | YLL012W (YEH1)        | 2      |

## 5 The measurement plate ✗

Seventy strains on one 384-well layout, five wells each, one pick per strain.

**Table 5:** What goes on the measurement plate. 70 strains at 5 wells each, one 384-well layout, one pick per strain. Wells per strain is the whole number of times 70 strains divide the 358 wells left after 6 are kept empty to fix the plate’s orientation and 20 are given to the wild-type reference, which gives 5. Every existing double is re-measured alongside the new strains, at no cost in construction work or in wells per strain.

| on the plate                     | strains   | wells      |
|----------------------------------|-----------|------------|
| singles, already built           | 11        | 55         |
| doubles, already built (Table 2) | 13        | 65         |
| doubles, to construct (Table 3)  | 25        | 125        |
| triples, to construct (Table 4)  | 20        | 100        |
| wild type, BY4741                | 1         | 20         |
| <b>total</b>                     | <b>70</b> | <b>365</b> |
| orientation blanks               | –         | 6          |
| spare                            | –         | 13         |
| <b>plate</b>                     |           | <b>384</b> |